
Glaciers Meet Ancient Capitals: Multi-Objective Optimization in Sustainable Tourism

Summary

With the rapid expansion of global tourism, overtourism has become a significant challenge for many popular tourist cities. As the capital of Alaska, Juneau enjoys substantial economic benefits from tourism but also faces increasingly severe environmental and social issues.

This paper addresses overtourism in Juneau by developing a multi-objective optimization model to maximize net tourism revenue (economic), minimize glacier retreat (environmental), and maximize resident satisfaction (social). The model is solved using the **Non-dominated Sorting Genetic Algorithm II (NSGA-II)**, with the optimal solution selected from the **Pareto front** via the **Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS)**. The results yield two optimal scenarios: (1) **1,525,294 tourists** with tax allocations of **23%** for environmental protection, **5%** for infrastructure, and **72%** for community development; (2) **1,405,717 tourists** with tax allocations of **22%** for environmental protection, **5%** for infrastructure, and **72%** for community development. These solutions balance economic, environmental, and social objectives.

To assess the model's generalizability, it is applied to overtourism-affected cities, with Beijing as a case study. Adapted to Beijing's historical and cultural context, the model integrates localized factors like cultural landscape maintenance costs. Utilizing NSGA-II, it determines optimal tourist numbers and tax allocation, showcasing its adaptability and robustness across diverse urban settings.

To further explore the impact of scenic location on optimization strategies, this paper introduces a **carbon emission quantification framework based on tourist travel choices**. By integrating key variables such as tourist numbers, per capital carbon emission coefficients, and travel distances, the model precisely calculates carbon emissions for each scenic site and constructs a multi-objective optimization function to balance economic revenue, environmental protection, social well-being, and carbon emission minimization. Through **K-Means clustering analysis**, the model categorizes scenic sites based on their objective weight vectors and proposes targeted optimization strategies for each category.

This paper optimizes tourist distribution between popular and less popular sites using the **Gini Index** to measure balance and a **subsidy mechanism** to incentivize visits to less popular sites via government-funded fee reductions. The model applies dynamic constraints to keep tourist numbers at popular sites within carrying capacity while ensuring minimum thresholds at less popular sites. Using NSGA-II and **Particle Swarm Optimization (PSO)**, it derives an optimized distribution scheme for more equitable tourism resource allocation.

In conclusion, this paper proposes a framework for the sustainable development of Juneau's tourism industry, providing scientific insights and policy recommendations. The model recommends maintaining annual tourist numbers **between 1,400,000 and 1,500,000** and allocating over **70%** of tax revenue to community development and environmental protection, thereby improving resident satisfaction and mitigating environmental degradation.

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1 Introduction

1.1 Problem Background

Located near the coastline of southeastern Alaska, Juneau is a city known for its natural beauty. As the capital of Alaska, Juneau is rich in tourism resources, including spectacular glaciers, dense rainforests and abundant wildlife resources. These unique natural landscapes attract a large number of tourists, especially cruise ship travelers.

However, with the growth of the global economy and the surge in the number of tourists, the City of Juneau is facing a serious challenge of over-tourism. Excessive tourists are placing tremendous pressure on the environment, infrastructure, and quality of life for residents of the City of Juneau. In particular, the problem of glacier retreat caused by over-tourism has sparked widespread local concern about the sustainability of the tourism industry.



Figure 1: Mendenhall Glacier

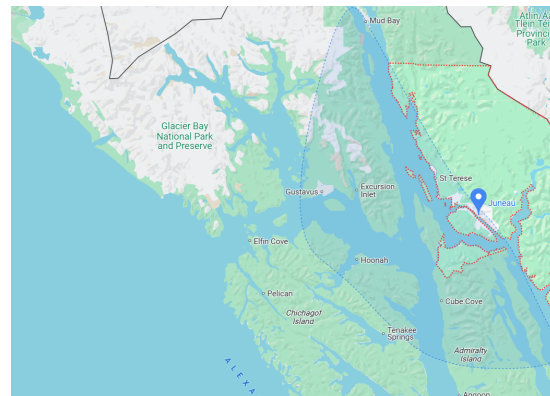


Figure 2: Location of Juneau

1.2 Clarifications and Restatements

The objective of this study is to establish a sustainable tourism industry model that can address the challenges of tourism development in the context of limited environmental carrying capacity in the city of Juneau. The model's core objective is to comprehensively maximise economic, environmental and social benefits through a multi-objective optimisation approach, and to improve investment efficiency through in-depth analysis of the optimal allocation of tourism revenue. The subsequent phase of the study will involve the extension of this model to a wider region, with the aim of providing more generalised solutions for sustainable development in different scenarios. The specific research steps are as follows:

1. **Model Building:** A multi-objective optimisation framework is utilised to construct a comprehensive model that can concurrently evaluate economic, environmental and social benefits with a view to achieving the goal of synergistic optimisation of the three.
2. **Reinvestment of Income:** A dynamic feedback mechanism is introduced to reinvest a fixed proportion of tourism revenue to form a sustainable capital cycle, thereby supporting long-term environmental protection and social welfare improvement.

3. **Sensitivity Analysis:** The Sobol global sensitivity analysis method is utilised to identify key parameters that significantly affect the model output results. Consequently, targeted optimisation strategies are formulated to achieve efficient resource allocation.
4. **Model Universality Verification:** Based on model construction and optimisation, the model is further applied to other cities affected by overtourism or with a small number of tourists to verify its applicability and robustness in different situations.

1.3 Our Work

Our solution mindmap is shown below.

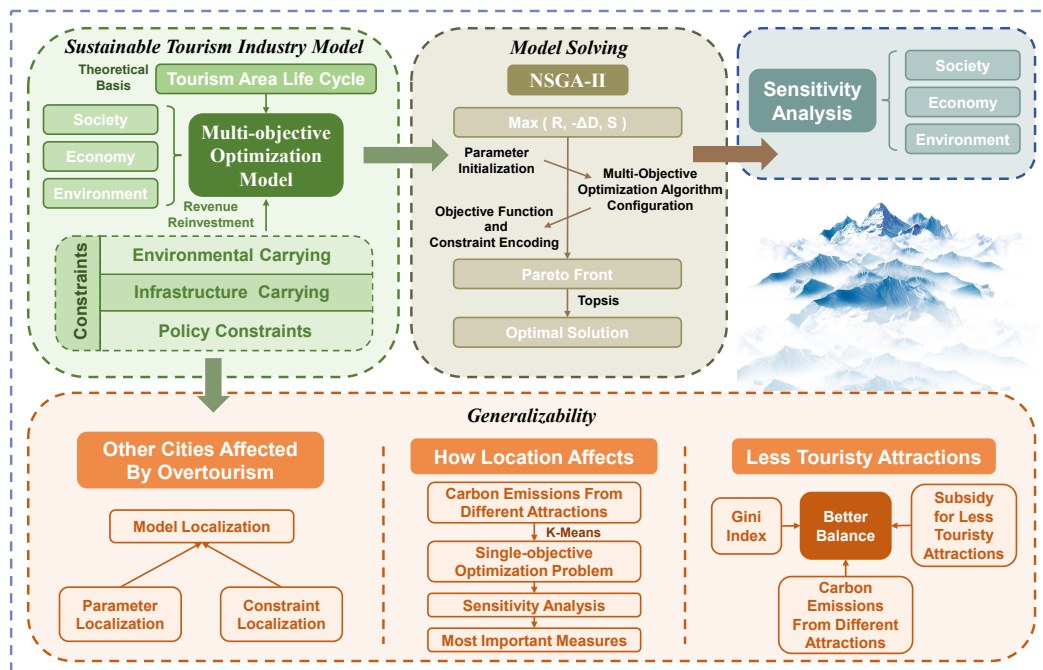


Figure 3: Flow Chart of Our Work

2 Model Preparation

2.1 Assumptions and Justifications

Considering that practical problems always contain many complex factors, first of all, we need to make reasonable assumptions to simplify the model, and each hypothesis is closely followed by its corresponding explanation:

- **Assumption1:** The number of tourists is a continuous variable.
Justification: Due to the large number of tourists, the assumption that the number of tourists is a continuous variable does not affect the accuracy of the results.
- **Assumption2:** The consumption per tourist is fixed.
Justification: Assuming that each tourist's consumption is fixed can avoid the uncertainty caused by individual consumption differences, thus making the model more focused on the core issue.

- **Assumption3:** An increase in the number of tourists will accelerate environmental degradation.
Justification: An increase in the number of tourists will directly or indirectly lead to an increase in carbon emissions, which will accelerate environmental degradation. In Juneo City, this is reflected in the melting of the glaciers.
- **Assumption4:** The data we use is true and reliable.
Justification: Our data sources include international statistical agencies, academic databases, and Administrative Organ. Therefore, we can confidently state that the data we use is accurate and reliable for modeling purposes.

Additional assumptions are made to simplify analysis for individual sections. These assumptions will be discussed at the appropriate locations.

2.2 Notations

This chapter introduces the symbols and their descriptions used in the article, providing clear definitions and explanations to ensure consistency and ease of reference throughout the text.

Table 1: Notations

Symbol	Description
R	Net tourism revenue
P	Per capita tourist spending
V	Annual tourist volume
S	Resident satisfaction
e	carbon emission coefficient
ΔG	Glacier retreat rate
C	Cost
θ	Tax reinvestment ratio
N_{res}	Resident population

2.3 Data Collection

This article is based on the following table of data sources, which provides detailed information about the data source and their websites. The table ensures transparency and reproducibility by clearly outlining the key resources that support the research findings.

Table 2: Data Source and Websites

Data Source	Websites
NSIDC	https://nsidc.org/data/search#keywords=Mendenhall+Glacier/
ICCT	https://theicct.org/marine-cruising-flying-may22/
USGS	https://earthexplorer.usgs.gov/
CBJ	https://juneau.org/
DOT&PF	https://dot.alaska.gov/
Booking.com	https://www.booking.com/
BMBS	https://tjj.beijing.gov.cn/
CNRDS	https://www.cnrds.com/Home/Login#/

3 Model Establishment: Multi-Objective Optimization

3.1 Tourism Area Life Cycle

According to Butler's (1980) destination life cycle theory[1], the evolution of a tourist destination typically goes through six phases: exploration, involvement, development, consolidation, stagnation, and decline (or revitalization). Based on the current situation analysis of the city of Juneau, the city may be in the consolidation or stagnation stage.

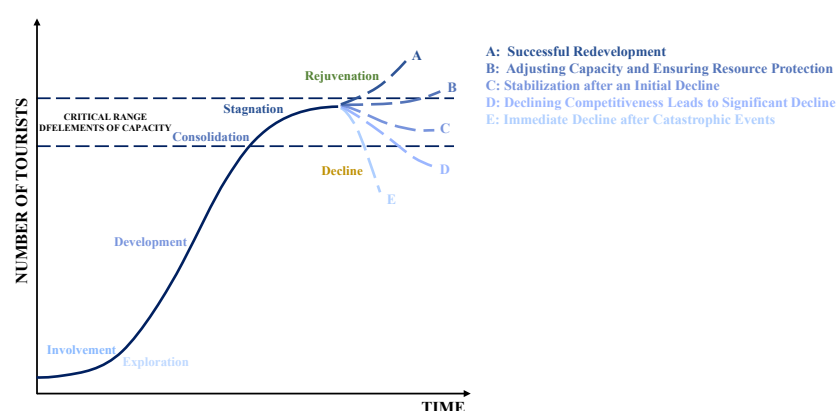


Figure 4: Hypothetical Evolution of a Tourist Area

In the development and consolidation stages, rapid growth in visitor numbers often leads to significant environmental pressures. The accelerated retreat of the Mendenhall Glacier in Juneau, for example, is not only related to climate change, but also closely linked to excessive tourism activities. In addition, as tourism continues to expand, local residents' attitudes towards tourism may gradually change from early support to dissatisfaction and even resistance. This social dynamic change further confirms the typical characteristics of the consolidation and stagnation stages. To prevent Juneau from entering the decline stage, it is urgent to achieve a balance between economic, environmental and social goals through sustainable tourism strategies.

3.2 A Sustainable Model Based on Multi-Objective Optimisation

The increase in the number of tourists in this region has had significant economic benefits, but it has also triggered negative impacts such as environmental degradation and reduced resident satisfaction. Thus, this group has chosen to use a Multi-Objective Optimization Model to systematically analyze this issue. The incorporation of tax revenue reinvestment in environmental protection, infrastructure enhancement, and community development initiatives within the model, manifested as a dynamic feedback mechanism, facilitates a more comprehensive and scientific evaluation of the issue. This, in turn, provides a theoretical foundation for the formulation of sustainable tourism policies.

3.2.1 Objective Function

1.Economic Objective:

In the context of sustainable tourism management, the overarching economic objective is to optimise the net benefits of tourism while ensuring its long-term sustainability. Consequently, the economic imperative is to optimise the financial gains derived from tourism by effectively managing the number of tourists and the distribution of taxes, while allocating funds towards environmental protection and community development. Therefore, we do the following calculations:

The total annual tourist expenditure is equal to:

$$\text{Tourist Expenditure} = P \times V \quad (1)$$

Cost of infrastructure damage caused by tourists is equal to:

$$C_{\text{infra}} = a_1 V + b_1 \quad (2)$$

Cost of environmental damage caused by tourists is equal to:

$$C_{\text{env}} = a_2 V + b_2 \quad (3)$$

And Tax allocation for infrastructure is equal to:

$$\theta_{\text{infra}} = \frac{C_{\text{env}}}{R} \quad (4)$$

Substituting $\theta_{\text{infra}} = \frac{C_{\text{env}}}{R}$ back into the profit function $R(V, \theta) = P \times V - C_{\text{infra}} - C_{\text{env}}$, we obtain the final profit function:

$$R(V, \theta) = \frac{P \times V - C_{\text{env}}(V)}{1 + \theta_{\text{infra}}} \quad (5)$$

Based on the Juno City Government's financial reports over the years and data on the number of tourists, this study used least squares regression analysis to obtain the following coefficient estimates:

$$\begin{aligned} \text{Const} &= -1.840140 \times 10^6 \\ x_1 &= -1.867107 \times 10^{-3} \\ x_2 &= 1.720156 \times 10^{-1} \end{aligned}$$

2.Environmental Objective:

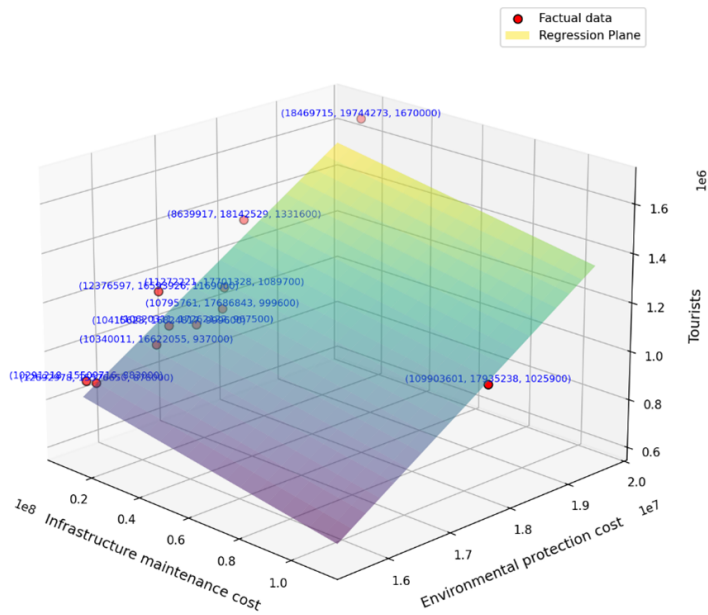


Figure 5: Three-variable Regression Plane for the Number of Tourists and Cost Variables

In the context of sustainable tourism management, the core of environmental objectives is to achieve a balance between the development of tourism and the protection of natural resources. The retreat of the Mendenhall Glacier in Juneau poses a threat not only to the local ecosystem, but also to the long-term sustainability of the tourism industry.

Thus, we construct the environmental objective function as:

$$\text{Min } \Delta G(V, \theta) = k_1 V + k_2(c \cdot V) - \beta(\theta_{\text{env}} R) \quad (6)$$

where:

- k_2 is the fitted parameter, Fitted $k_2 = -0.002650754445271545$.
- $c = 0.756$ is the carbon emission coefficient.
- $\beta = 0.1$ is the impact coefficient of environmental investment on glacier retreat.
- θ_{env} is the tax allocation ratio for environmental protection.

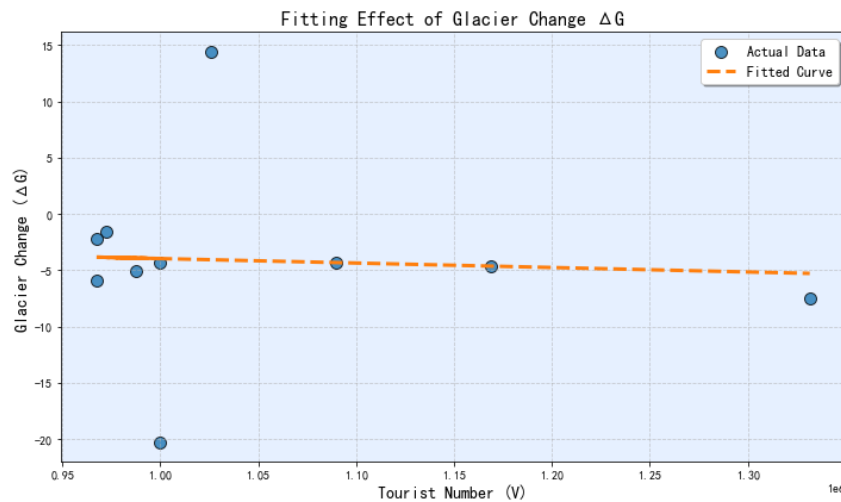


Figure 6: Fitting Effect of Glacier Change

3.Social Objective

In sustainable tourism management, the core of the social objective is to maximize resident satisfaction, ensuring that the healthy development of the tourism industry does not negatively impact the local community. The rapid growth of tourism in Juneau, while bringing significant economic benefits, has also led to resident dissatisfaction due to a decline in quality of life. Therefore, the construction of the social objective aims to optimize tourist numbers and tax allocation, enhance residents' acceptance and satisfaction with the tourism industry, and achieve a balance between sustainable tourism development and community well-being.

Based on the relevant data from the City of Juneau government reports, multiple fittings were conducted.

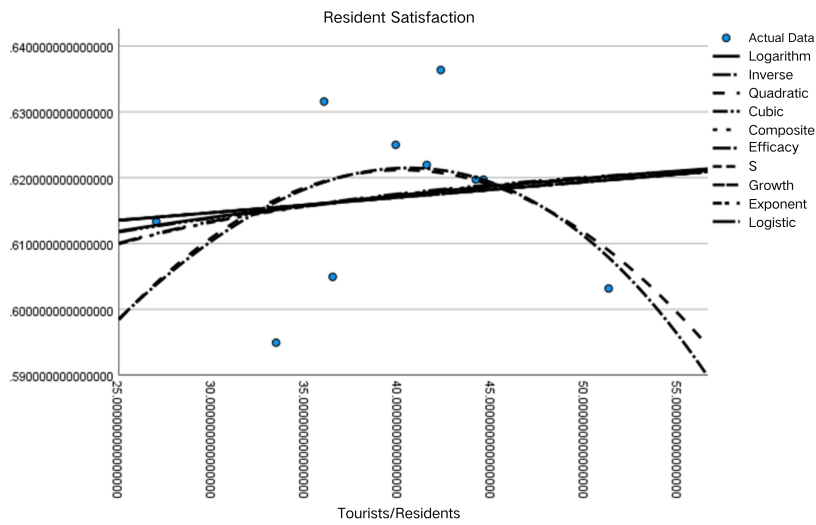


Figure 7: Summary of Resident Satisfaction Fitting Methods

The optimal fit was ultimately determined to be a quadratic regression. The expres-

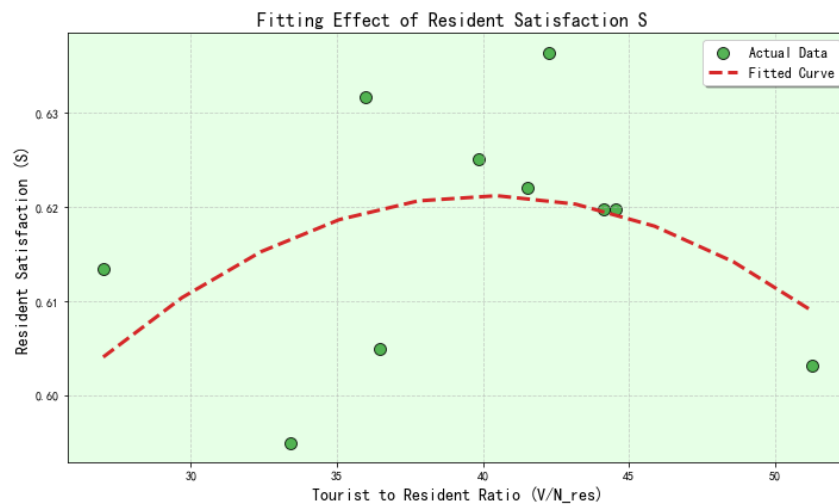


Figure 8: Fitting Effect of Resident Satisfaction S

sion for resident satisfaction (S) is obtained as:

$$\text{Max } S = S_0 - \alpha \cdot \left(\frac{V}{N_{\text{res}}(V)} \right) + \sigma \cdot \left(\frac{V}{N_{\text{res}}(V)} \right)^2 + \gamma \cdot (\theta_{\text{comm}} \cdot R) \quad (7)$$

where,

- S_0, α, σ are parameters obtained from fitting, Fitted $S_0=0.461432436731945$, $\alpha=-0.007955985703703043$, $\sigma=-9.90620996289211e-05$.
- $N_{\text{res}}(V)$ is the number of residents, related to the number of tourists V .
- γ is the impact coefficient of community investment on resident satisfaction = 0.2.
- θ_{comm} is the tax allocation for community investment.

3.2.2 Constraint

1.The Upper Limit of Environmental Carrying Capacity V_{env}

For the upper limit of environmental carrying capacity, based on relevant data, our team believes that the annual glacier retreat limit ΔG_{max} should not exceed 3 meters. Based on this, the environmental carrying capacity constraint can be expressed as:

$$0 \leq V \leq V_{\text{env}} = \frac{\Delta G_{\text{max}}}{k_1} \quad (8)$$

where V is the number of tourists, V_{env} is the upper limit of environmental carrying capacity and k_1 is the conversion coefficient between glacier retreat and tourist numbers.

2.The Upper Limit of Infrastructure Carrying Capacity V_{infra}

In determining the upper limit of infrastructure capacity, the team gave full consideration to the port capacity (HC), traffic capacity (TC) and accommodation capacity (AC) of the city of Juneau. The specific data are as follows: Then we obtain the constraints:

$$0 \leq V \leq V_{\text{infra}} = \min(\text{HC}, \text{TC}, \text{AC}) \quad (9)$$

3.Policy Constraints V_{daily}

According to the policy on the upper limit of the number of tourists in 2026 announced by the Juno municipal government, the following restrictions on the daily number of tourists have been set:

$$\begin{cases} 0 \leq V_{\text{daily}} \leq 12000, & \text{Saturday} \\ 0 \leq V_{\text{daily}} \leq 16000, & \text{Non-Saturday} \end{cases} \quad (10)$$

	Daily Capacity Limit		Annual Capacity Limit	Data Source
HC	Saturday	Non-Saturday	2376000	CBJ
	12000	16000		
TC	15000		2310000	DOT&PF University of Alaska
AC	-		3650000	Booking.com

In summary, we obtain the overall constraints:

$$\begin{cases} 0 \leq V \leq V_{\text{env}} = \frac{\Delta G_{\text{max}}}{k_1} \\ 0 \leq V \leq V_{\text{infra}} = \min(\text{HC}, \text{TC}, \text{AC}) \\ \begin{cases} 0 \leq V_{\text{daily}} \leq 12000, & \text{Saturday} \\ 0 \leq V_{\text{daily}} \leq 16000, & \text{else} \end{cases} \end{cases}$$

$$\begin{cases} \theta_{\text{env}} + \theta_{\text{infra}} + \theta_{\text{comm}} = 1 \\ 0 \leq \theta_{\text{env}}, \theta_{\text{infra}}, \theta_{\text{comm}} \leq 1 \end{cases}$$

3.3 Model Solving: NSGA-II

3.3.1 Solving Method Selection

For the multi-objective optimization model we constructed, our team innovatively adopted the Non-dominated Sorting Genetic Algorithm II (NSGA-II), instead of the traditional linear solution method[2].

3.3.2 Model Solving

The multi-objective problem can be represented as:

$$\text{Max}(R, -\Delta G, S) \quad (11)$$

Solving Procedure

Step 1: Parameter Initialization

Set model parameters:

$k_i = 0.002$ meters/thousand people, $\alpha = 0.5$, $\beta = 0.1$ meters/million USD, etc.

Define variable ranges $V \in [0, 2.31 \times 10^6]$, $\theta \in [0, 1]$. Where:

- R is the economic objective (maximize).
- $-\Delta G$ is the environmental objective (minimize glacier retreat rate, with the negative change in glacier size as the maximum optimization problem).
- S is the social objective (maximize resident satisfaction).

Step 2: Multi-Objective Optimization Algorithm Configuration

- Population size: 100 (balancing computational efficiency and diversity).
- Number of iterations: 500 (ensuring convergence).
- Crossover probability: 0.9, mutation probability: 0.1 (maintaining population diversity).

Step 3: Objective Function and Constraint Encoding

Transform the objective functions $R, \Delta G, S$ into fitness functions, and handle constraints using the penalty function method.

$$\text{Penalty Function} = \lambda \cdot \max(0, V - V_{\text{max}})^2 + \mu \cdot |\theta_{\text{env}} + \theta_{\text{infra}} + \theta_{\text{comm}} - 1| \quad (12)$$

where λ and μ are penalty coefficients. **Step 4: Iterative Solution and Pareto Front Generation**

The algorithm outputs a set of non-dominated solutions (Pareto optimal solutions), each corresponding to a set $(V, \theta_{\text{env}}, \theta_{\text{infra}}, \theta_{\text{comm}})$. **Step 5: Solution Selection and Trade-off Analysis**

Select the final solution from the Pareto front using TOPSIS (Technique for Order of Preference by Similarity to Ideal Solution). Then we get two optimal solutions.

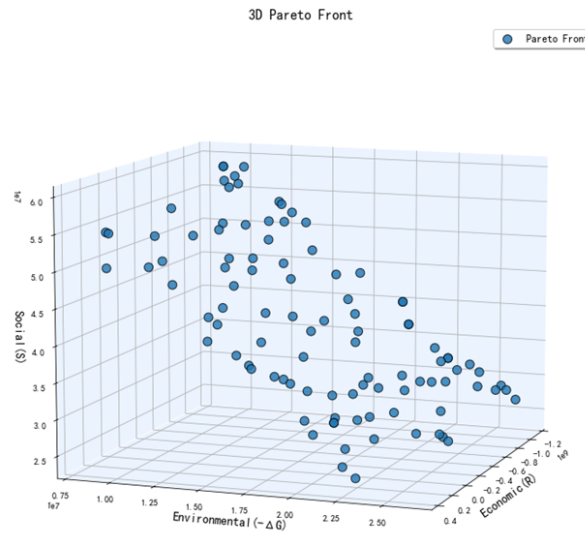


Figure 9: 3D Pareto Front: Social, Environment and Economic

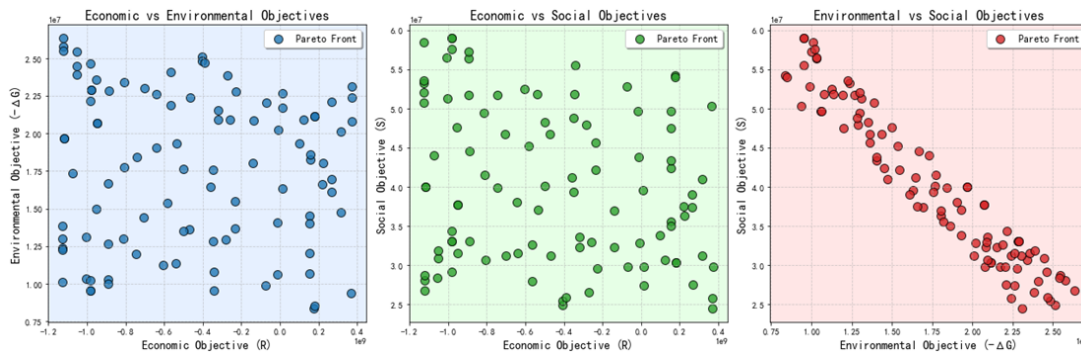


Figure 10: Set of Pareto Front Diagrams

- **Solution1:** $V = 1525294, \theta = (0.23, 0.05, 0.72)$
Target Value: $R = -982249868.41, -\Delta G = 9529678.42, S = 58948041.05$
- **Solution2:** $V = 1405717, \theta = (0.22, 0.05, 0.72)$
Target Value: $R = 177089345.66, -\Delta G = 8344323.61, S = 54178986.08$

3.3.3 Sensitivity Analysis

Sensitivity analysis quantifies the impact of input parameters on the output objectives, helping us understand the dynamic behavior of the model. In this study, we use the Sobol sensitivity analysis method, which is based on variance decomposition, to analyze the impact of different input parameters and their interactions.

The analysis results are as follows: 1. Economic Objective R :

- The first-order sensitivity index for V (tourist number) is 0.861, and the total sensitivity index is 0.901, indicating that the number of tourists has the most significant impact on the economic objective.
- The first-order sensitivity index for θ_{infra} (tax allocation for infrastructure construction) is 0.095, and the total sensitivity index is 0.134, indicating that infrastructure tax ratio has a certain impact on the economic objective.
- The first-order sensitivity index for θ_{env} (tax allocation for environmental protection) and θ_{comm} (community tax ratio) is 0, indicating that these factors have little effect on the economic objective.

2. Environmental Objective ΔG :

- The first-order sensitivity index for V (tourist number) is 0.400, and the total sensitivity index is 0.564, indicating that the number of tourists has a significant impact on the environmental objective.
- The first-order sensitivity index for θ_{env} (tax allocation for environmental protection) is 0.376, and the total sensitivity index is 0.528, indicating that environmental tax ratio has a significant impact on the environmental objective.
- The first-order sensitivity index for θ_{infra} (tax allocation for infrastructure construction) is 0.047, and the total sensitivity index is 0.084, indicating that infrastructure tax ratio has a relatively small impact on the environmental objective.
- The first-order sensitivity index for θ_{comm} (tax allocation for community investment) is 0, indicating that it has no impact on the environmental objective.

3. Social Objective S :

- The first-order sensitivity index for V (tourist number) is 0.398, and the total sensitivity index is 0.555, indicating that the number of tourists has a significant impact on the social objective.
- The first-order sensitivity index for θ_{comm} (tax allocation for community investment) is 0.374, and the total sensitivity index is 0.524, indicating that the community tax ratio also significantly affects the social objective.
- The first-order sensitivity index for θ_{infra} (tax allocation for infrastructure construction) is 0.042, and the total sensitivity index is 0.084, indicating that the infrastructure tax ratio has a relatively small impact on the social objective.
- The first-order sensitivity index for θ_{env} (tax allocation for environmental protection) is 0, and the total sensitivity index is 0, indicating that it has no impact on the social objective.

4 Model Promotion and Solving

In the initial phase of the study, a multi-objective optimization model was developed to ensure the sustainability of tourism in Juneau, Alaska. This model meticulously evaluates the objectives across three dimensions: maximizing economic benefits, minimizing ecological damage, and maximizing social and resident satisfaction.

The model's utility lies not only in its capacity to provide scientific policy recommendations for Juneau but also in its potential to serve as a viable reference point for other regions grappling with the challenges posed by overtourism. By recognizing the distinct challenges faced by various cities in the realm of sustainable tourism development and adapting the model's parameters to local contexts, the unique challenges associated with sustainable development in different cities can be effectively addressed.

4.1 Model Promotion: Popular Tourist Areas

As one of China's most significant tourist cities, Beijing has witnessed a marked rise in overtourism in recent years. Research conducted by the tourism consultancy IPK International has identified Beijing as one of the cities most impacted by this phenomenon. The primary challenges posed by this phenomenon in Beijing can be categorized as follows:

1. Challenges to Beijing's infrastructure, such as the cost of traffic congestion and subway maintenance
2. The environmental indicators in Beijing, including air quality and carbon dioxide emissions, have been significantly influenced.
3. Increased maintenance costs for the Beijing municipal government to protect ancient buildings and cultural relics.

This group has localized and modified the model based on the above issues.

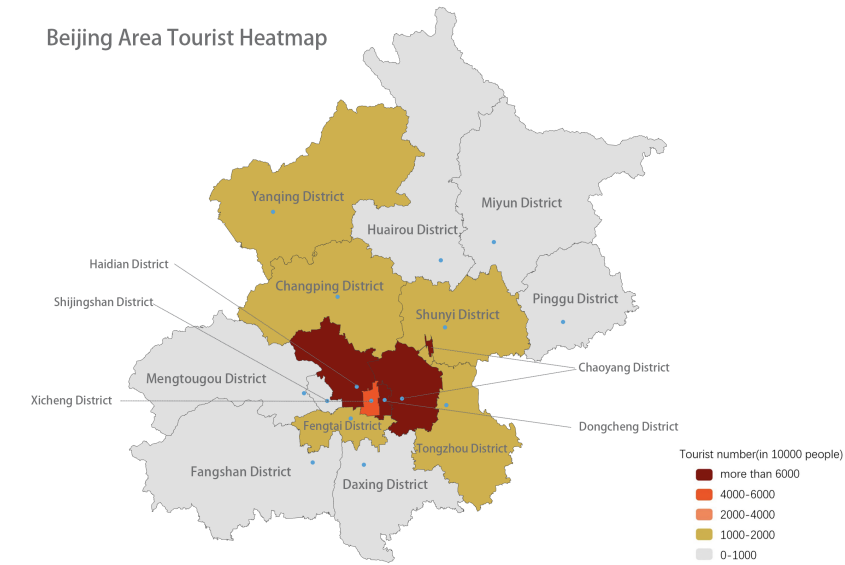


Figure 11: Beijing Area Tourist Heatmap

4.1.1 Model Localization

1. Economic Objective

Given Beijing's extensive historical background, it is inevitable that tourists will damage the floor tiles and cultural relics at attractions during their visits. Therefore, based on the first-order model, we introduce the cost, C_{culture} , of damage caused by tourists to the historical and cultural landscape of Beijing.

We obtained the expenditure data for the maintenance of ancient buildings and the protection of cultural relics based on relevant data, and used the least squares method to regress the expenditure equation for ancient building maintenance. Ultimately, we derived the localized and adjusted economic objective function:

$$\text{Max } R(V, \theta) = P \times V - C_{\text{infra}} - C_{\text{env}} - C_{\text{culture}} \quad (13)$$

where:

- $P \times V$ is the tourism revenue,
- C_{infra} is the infrastructure maintenance cost,
- C_{env} is the environmental protection cost,
- C_{culture} is the cultural preservation cost.

2. Environmental Objective

In the quantification of environmental objectives, given the significant geographical differences between Beijing and Juneau (with a large latitude and longitude span), the ecological characteristics of the two cities exhibit notable disparities, leading to distinct environmental challenges posed by tourism. This study categorizes the environmental impact of tourists in Beijing into two types: damage to historical relics and damage to the ecological environment.

For the historical cultural relics damage index k_1 , we derived it based on relevant data using the least squares regression method.

Regarding ecological environmental damage, we calculated the average carbon emissions per tourist and, based on related research, determined the carbon treatment cost coefficient k_2 .

Finally, by calculating the theoretical environmental loss value without government subsidies, this study derives the environmental benefit β per unit of environmental investment, thereby constructing the environmental objective function for Beijing:

$$\text{Min } \Delta E(V, \theta) = k_1 V + k_2(\text{CO}_2) - \beta(\theta_{\text{env}} R) \quad (14)$$

where $k_1 = 4.58$, $k_2 = 261$, $\text{CO}_2 = 34.1V$, $\beta = 0.1$

To calculate the environmental benefit per unit of environmental investment (β) in Beijing, it is necessary to first estimate the theoretical environmental loss value without government subsidies. The steps are as follows:

A. Definition of Theoretical Environmental Loss Value

The theoretical environmental loss value refers to the economic loss caused by environmental pollution in the absence of any environmental protection measures. The environmental cost C can be expressed as:

$$C = C_d + C_{uA} \quad (15)$$

where C_d represents direct environmental costs (e.g., pollution control and remediation costs), and C_{id} represents losses to other sectors (e.g., health and ecological losses).

B. Data Collection

According to data from the Beijing Municipal Finance Bureau, the recent environmental pollution control investment is 21,958.63 million yuan (including government subsidies).

C. Calculation of Environmental Investment Without Subsidies

Assuming government subsidies account for 30% of the total investment, the environmental investment without subsidies is:

$$\text{Environmental Investment Without Subsidies} = 21,958.63 \times \frac{1}{1 - 0.3} = 31,369.47 \quad (16)$$

D. Estimation of Theoretical Environmental Loss Value

Assuming the theoretical environmental loss value without subsidies is 50% of the environmental investment, then:

$$\text{Theoretical Environmental Loss Value} = 31,369.47 \times 0.5 = 15,684.74 \quad (17)$$

E. Calculation of Environmental Benefit per Unit Investment (β)

The environmental benefit β represents the environmental benefit per unit of investment, calculated as:

$$\beta = \frac{\text{Theoretical Environmental Loss Value}}{\text{Environmental Investment Without Subsidies}} = \frac{15,684.74}{31,369.47} \approx 0.5 \quad (18)$$

The results indicate that the environmental benefit per unit of investment (β) in Beijing is approximately 0.5, meaning that for every 1 yuan invested in environmental protection, 0.5 yuan of environmental benefit is generated.

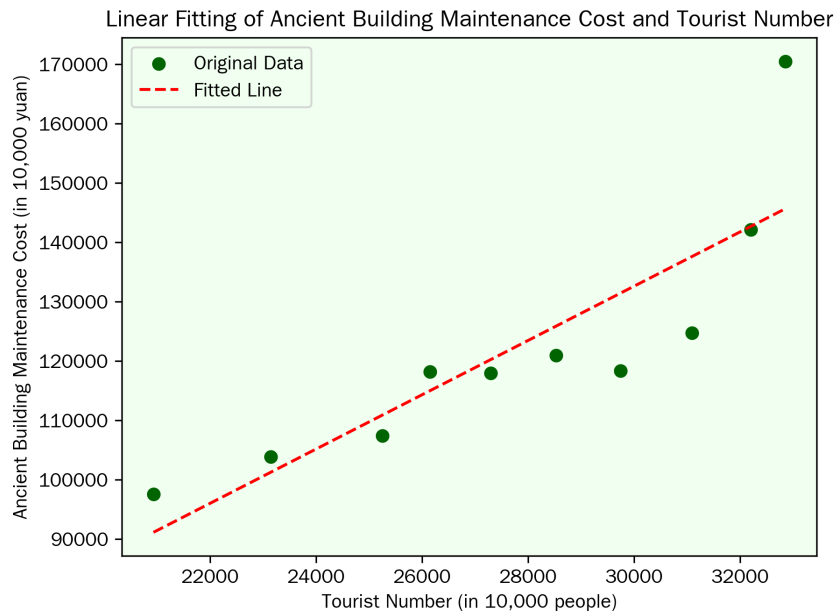


Figure 12: Linear Relationship Between Tourist Numbers and Ancient Building Maintenance Costs

3. Social Objective

Our team adopted the model from the previous paper and obtained the baseline satisfaction S_0 through relevant surveys. By regressing the annual tourist numbers and resident population with the historical resident satisfaction data, we derived the resident satisfaction coefficient α . Based on the government-published community investment amounts and the corresponding resident satisfaction data, we calculated the gain coefficient γ for the impact of community investment on resident satisfaction.

Based on the above analysis, the social objective function for Beijing is constructed as follows:

$$S = S_0 - \alpha \cdot \left(\frac{V}{N_{\text{res}}(V)} \right) + \sigma \cdot \left(\frac{V}{N_{\text{res}}(V)} \right)^2 + \gamma \cdot (\theta_{\text{comm}} \cdot R) \quad (19)$$

where $S_0 = 48.7791$, $\alpha = 7.1898$, and $\sigma = 0.3395$.

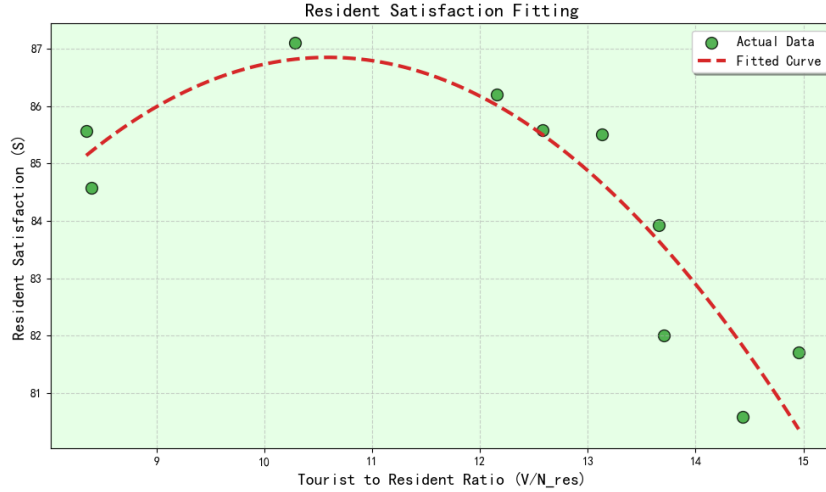


Figure 13: Fitting Effect of Resident Satisfaction

4.1.2 Localization of Constraints

Based on the specific conditions of Beijing, our team adjusted the constraints by collecting relevant data and derived the overall constraint conditions as:

$$\begin{cases} 0 \leq V \leq V_{\text{env}} \\ 0 \leq V \leq V_{\text{infra}} = \min(\text{TC}, \text{AC}) \\ 0 \leq V \leq V_{\text{culture}} \end{cases} \quad (20)$$

$$\begin{cases} \theta_{\text{env}} + \theta_{\text{infra}} + \theta_{\text{comm}} \leq 1 \\ 0 \leq \theta_{\text{env}}, \theta_{\text{infra}}, \theta_{\text{comm}} \leq 1 \end{cases} \quad (21)$$

4.1.3 Model Solution: NSGA-II

Building upon the solution method outlined in the first section, this study employs the NSGA-II for multi-objective optimization, iteratively generating the Pareto Front. Subsequently, the TOPSIS method is applied to select the optimal solution from the Pareto Front. The final results are as follows:

$$V = 285, 237, 246 \quad \theta = (0.62, 0.01, 0.37) \quad (22)$$

The corresponding objective function values are:

$$R = 5, 556.24379298 \quad \Delta E = 26, 659, 335, 208.98 \quad S = 963, 844.54 \quad (23)$$

4.2 Model Extension: Impact of Scenic Location

To quantify the environmental impact of different attraction choices, this study introduces a carbon emission quantification framework for attraction locations into the model.

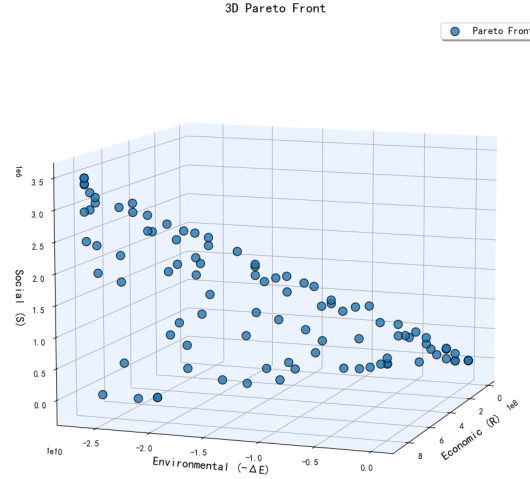


Figure 14: 3D Pareto Front: Social, Environment and Economic

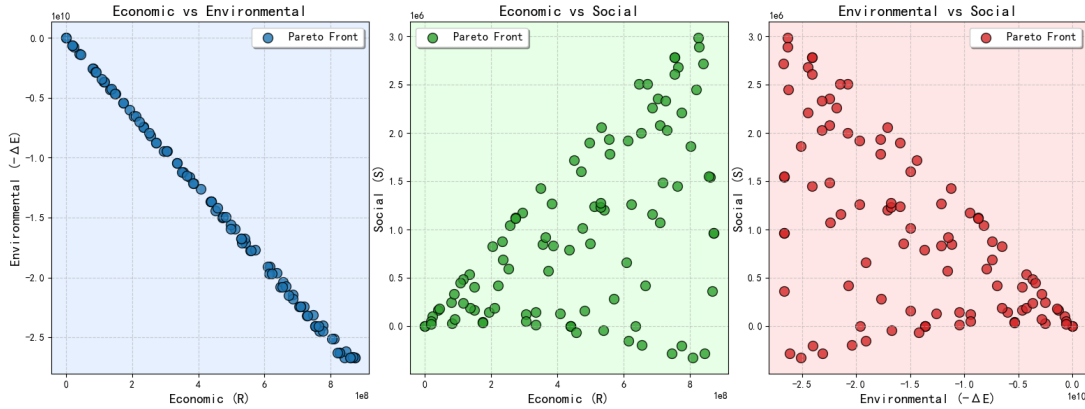


Figure 15: Set of Pareto Front Diagrams

Using Beijing as a case study, this research delves into how the model dynamically balances and optimizes the priorities of multiple objectives, such as economic and environmental goals, based on the geographical differences of scenic sites.

4.2.1 Model Update: Location-Based Modeling

This study adapts the Juneau City model to the context of Beijing, constructing a multi-objective optimization function for Beijing:

$$\begin{cases} \max R = P \times V - C_{\text{infra}} - C_{\text{env}} - C_{\text{culture}} \\ \min \Delta E(V, \theta) = k_1 V + k_2 (\text{CO}_2) - \beta(\theta_{\text{env}} R) \\ \max S(V, \theta) = S_0 - \alpha \left(\frac{V}{N_{\text{res}}} \right) + \sigma \left(\frac{V}{N_{\text{res}}} \right)^2 + \gamma(\theta_{\text{comm}} R) \end{cases} \quad (24)$$

The CO_2 mentioned here does not include emissions from transportation.

To fully account for the impact of scenic locations on carbon emissions from tourist travel, this study introduces a new objective function, the scenic CO_2 emission target[3]:

$$\Delta \text{CO}_{2i} = V_i \cdot \delta_i \cdot d_i \quad (25)$$

where:

- V_i denotes the number of tourists at the i -th scenic site.
- δ_i represents the per capita carbon emission coefficient at the i -th scenic site.
- d_i is the average distance from the i -th scenic site to tourists' residential locations.

Based on Beijing's annual carbon budget, this study further introduces a new constraint:

$$\sum_{i=1}^N \Delta \text{CO}_{2i} \leq B_{\text{carbon}} \quad (26)$$

4.2.2 Model Solution

Based on the previous content, this study constructs a multi-objective optimization function for scenic sites:

$$\text{Comprehension Objective}_i = \varepsilon \cdot R_i + \zeta \cdot (-\Delta E_i) + \eta \cdot S_i + \lambda \cdot (\Delta \text{CO}_{2i}) \quad (27)$$

The constraints are:

$$\begin{cases} 0 \leq V_i \leq V_{\text{env}_i} \\ 0 \leq V_i \leq V_{\text{infra}_i} \\ 0 \leq V \leq V_{\text{culture}} \end{cases} \quad (28)$$

$$\begin{cases} \sum_{i=1}^N \Delta \text{CO}_{2i} \leq B_{\text{carbon}} \\ \theta_{\text{env}} + \theta_{\text{infra}} + \theta_{\text{comm}} = 1 \\ 0 \leq \theta_{\text{env}}, \theta_{\text{infra}}, \theta_{\text{comm}} \leq 1 \end{cases} \quad (29)$$

Our team applied sensitivity analysis to obtain the specific values of ϵ , $\zeta + \lambda$ and η for 32 scenic spots in Beijing. Using K-Means clustering analysis, we categorized the attractions into three groups based on their carbon emission characteristics, economic benefits, and social impacts. This classification allows us to develop targeted improvement strategies and optimization measures for each category of attractions[4].

Scatter Plot of Attractions Clustering (Optimized, 3 Categories)

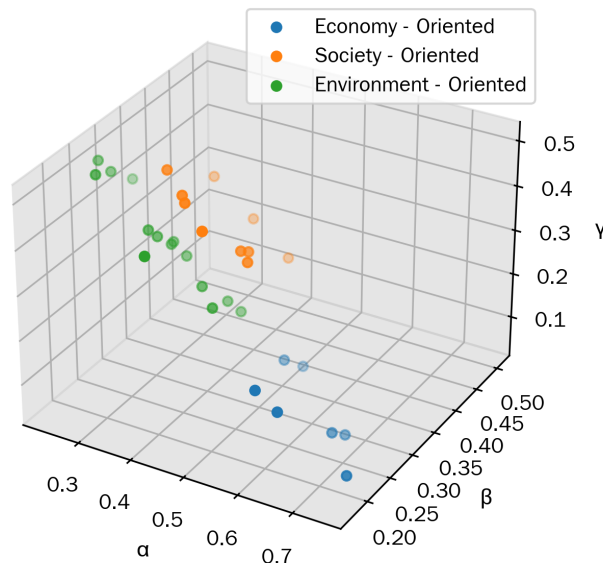


Figure 16: Scatter Plot of Attractions Clustering

4.2.3 Optimization Measures Selection

Based on the classification results, we propose the following optimization measures:

1. For attractions such as the Palace Museum and the Bird's Nest, where social factors have a greater impact, measures like visitor diversion can be implemented to reduce infrastructure pressure and resident dissatisfaction.
2. For attractions such as the Badaling Great Wall and Jingshan Park, where economic factors have a greater impact, direct bus services from urban areas can be introduced to enhance economic benefits.
3. For attractions such as the National Stadium and the Water Cube, where environmental factors have a greater impact, measures like strengthening environmental protection and optimizing visitor flow management can be adopted to improve sustainable development.

4.3 Model Extension: Less Popular Scenic Sites

4.3.1 Model Update: Gini Index

Building on the aforementioned model, this study further introduces an equilibrium indicator, dynamic constraints, and a subsidy mechanism for less popular scenic sites to regulate the distribution of tourists between popular and less popular sites, thereby achieving a more rational allocation of tourism resources.

1. Gini Index

The Gini index is a measure of inequality in distribution, which can be used to assess the balance of tourist distribution across different scenic sites. It provides a quantitative basis for governments and managers to evaluate the rationality of tourism resource allocation. The specific expression is as follows[5]:

$$D = 1 - \frac{\sum_{i=1}^N \sum_{j=1}^N |V_i - V_j|}{2N^2\bar{V}} \quad (30)$$

where:

- V_i denotes the number of tourists at the i -th scenic site,
- $\bar{V}$ represents the average number of tourists across all scenic sites,
- N is the total number of scenic sites.

The closer the equilibrium index D is to 1, the more balanced the tourist distribution; the closer it is to 0, the more unbalanced the distribution.

2. Dynamic Constraints

To optimize the distribution of tourists between popular and less popular scenic sites, this study establishes dynamic constraints for both types of sites as follows:

$$V_h \leq m \cdot V_{\text{cap},h} \quad V_c \geq n \cdot V_{\text{cap},c} \quad (31)$$

where:

- V_h denotes the actual number of tourists at popular scenic sites,
- $V_{cap,h}$ represents the maximum tourist capacity of popular scenic sites,
- m is the upper limit coefficient for tourists at popular scenic sites,
- V_c denotes the actual number of tourists at less popular scenic sites,
- $V_{cap,c}$ represents the maximum tourist capacity of less popular scenic sites,
- n is the lower limit coefficient for tourists at less popular scenic sites.

3.Subsidy Mechanism for Less Popular Tourist Attractions

To incentivize more tourists to visit less popular attractions, the government can implement a subsidy mechanism for entrance fees and other related costs. The subsidy cost for a less popular attraction can be expressed as follows:

$$C_{promoc} = k \cdot (V_{capc} - V_c) \quad (32)$$

where,

- C_{promoc} represents the government's subsidy cost for the less popular attraction.
- k denotes the subsidy coefficient for the less popular attraction.
- V_{capc} is the maximum visitor capacity of the less popular attraction.
- V_c is the actual number of visitors to the less popular attraction.

The economic objective function is subsequently updated to incorporate this subsidy mechanism:

$$R = P \times V - C_{infra} - C_{env} - C_{culture} - \sum_c C_{promoc} \quad (33)$$

4.Comprehensive Objective Function

Taking into account the various factors, we derive the comprehensive objective function as follows:

$$\text{Comprehensive Objective} = \varepsilon \cdot R + \zeta \cdot (-\Delta E) + \eta \cdot S + \lambda \cdot (-\Delta CO_2) + \mu \cdot D \quad (34)$$

subject to the following constraints:

$$\begin{cases} 0 \leq V_i \leq V_{env_i} \\ 0 \leq V_i \leq V_{infra_i} \\ 0 \leq V \leq V_{culture} \end{cases} \quad (35)$$

$$\begin{cases} \sum_{i=1}^N \Delta CO_{2i} \leq B_{carbon} \\ \theta_{env} + \theta_{infra} + \theta_{comm} = 1 \\ 0 \leq \theta_{env}, \theta_{infra}, \theta_{comm} \leq 1 \end{cases} \quad (36)$$

$$\begin{cases} V_h \leq m \cdot V_{cap,h} \\ V_c \geq n \cdot V_{cap,c} \end{cases} \quad (37)$$

4.3.2 Model Solution: Particle Swarm Optimization

The model aims to determine the specific values of ε , ζ , η , λ , and μ using NSGA-II, followed by the application of Particle Swarm Optimization (PSO) for parameter estimation. The detailed steps are as follows:

1. Particle Encoding: Each particle represents a potential solution, i.e., a distribu-

tion of visitors $V = [V_1, V_2, \dots, V_N]$, where V_i denotes the number of visitors at the i -th attraction.

2. Initialization of the Particle Swarm: Randomly generate a set of particles (visitor distributions) ensuring that all constraints are satisfied. Initialize the personal best solution B_{best} and the global best solution g_{best} .

3. Fitness Calculation: The fitness function is defined as the comprehensive objective function:

$$\text{Fitness} = \varepsilon \cdot R + \zeta \cdot (-\Delta E) + \eta \cdot S + \lambda \cdot (-\Delta CO_2) + \mu \cdot D \quad (38)$$

Calculate the fitness value for each particle.

4. Update Particle Position and Velocity: According to the PSO update rules:

$$v_i(t+1) = w \cdot v_i(t) + c_1 \cdot r_1 \cdot (p_{\text{best},i} - x_i(t)) + c_2 \cdot r_2 \cdot (g_{\text{best}} - x_i(t)) \quad (39)$$

$$x_i(t+1) = x_i(t) + v_i(t+1) \quad (40)$$

where:

- w is the inertia weight,
- c_1, c_2 are the learning factors,
- r_1, r_2 are random numbers.

5. Update Personal and Global Best Solutions: After each iteration, update the personal best position B_{best} for each particle. Update the global best position g_{best} based on the personal best positions of all particles until the termination criteria are met.

5 Result Analysis and Policy Recommendations

In the final solution, several scenarios may arise:

1. Concentration of Tourists in Popular Attractions: If the weights for popular attractions (e.g., ε, ζ) significantly outweigh those for less popular attractions (e.g., η, λ) during the optimization process, tourists will predominantly concentrate in popular attractions.

2. Balanced Distribution of Tourists: If the subsidy mechanism for less popular attractions is sufficiently incentivized (e.g., a high subsidy coefficient k) and the weight for the Gini index is substantial (μ is large), tourists will be more evenly distributed across various attractions.

3. Environmentally and Sustainably Oriented Tourist Distribution: If the weight for environmental goals (e.g., reducing carbon emissions) is significant (λ is high), the model may favor optimizing tourist travel modes and reducing the burden on attractions to maximize environmental benefits.

We propose the following policy recommendations:

1. For Overcrowded Popular Attractions: Implement stricter visitor flow restrictions, increase subsidies for less popular attractions, and enhance the appeal of less popular attractions (e.g., by adding cultural or natural experiences) to redistribute tourists and alleviate pressure on popular attractions.

2. Environmental Protection Measures: If the model favors environmental goals,

the government can adopt policies such as incentives for eco-friendly travel, restrictions on high-carbon emission transportation, and improvements in environmental facilities within attractions to ensure that tourism development aligns with environmental protection.

6 Model Evaluation

6.1 Strengths

1. **Universality and Scalability:** The model is not only applicable to the city of Juneau but has also been validated for its adaptability in other cities through localized adjustments, such as the case study in Beijing. This universality enables the model to provide tailored sustainable development solutions for diverse urban contexts.
2. **Flexible Constraints:** The model incorporates multiple constraints, including environmental carrying capacity, infrastructure capacity, and policy limitations. It dynamically adjusts tourist numbers and tax allocation schemes based on real-world conditions.

6.2 Weaknesses

1. **Data Dependency:** The accuracy and reliability of the model are highly dependent on the quality of the input data. Inaccurate or incomplete data may lead to optimization results that deviate from actual conditions.
2. **High Complexity:** The model involves multiple objective functions, constraints, and optimization algorithms, resulting in high computational complexity. This necessitates longer computation times and greater computational resources.

7 Memorandum

To: The Tourist Council of Juneau

From: Team #2517076

Subject: **Comprehensive Sustainable Tourism Development Framework and Policy Recommendations for Juneau**

Date: **January 27, 2025**

Dear Members of the Juneau Tourism Committee,

Our research team has developed a multi-objective optimization model to help Juneau balance economic growth, environmental protection, and community well-being. Below are the key findings and recommendations:

Model Predictions and Results

Our model, using NSGA-II, has produced two optimal solutions, each corresponding to different tourist numbers and tax allocation ratios:

1. Tourist numbers of 1,525,294, with tax allocation ratios of 23% for environmental protection, 5% for infrastructure, and 72% for community development.
2. Tourist numbers of 1,405,717, with tax allocation ratios of 22% for environmental protection, 5% for infrastructure, and 72% for community development.

Effectiveness of Measures

1. **Tourist Number Control:** We found that the number of tourists significantly impacts economic, environmental, and social goals. Appropriately reducing tourist numbers can effectively alleviate environmental pressure and improve resident satisfaction.
2. **Tax Reallocation:** Allocating a larger proportion of taxes to community development significantly enhances resident satisfaction, while environmental protection investments positively slow glacier retreat.

Optimization Recommendations

Based on the model results, we propose the following optimization recommendations:

1. **Tourist Number Regulation:** We recommend controlling annual tourist numbers between 1,400,000 and 1,500,000 to achieve a balance. This can be achieved through measures such as daily visitor limits.
2. **Tax Allocation Optimization:** We suggest allocating a larger proportion of taxes (over 70%) to community development and environmental protection to enhance resident satisfaction and slow environmental degradation.
3. **Environmental Protection Measures:** Increase investments in glacier protection and carbon emission control by implementing visitor flow restrictions to reduce direct pressure on glaciers.

We believe that by implementing these measures, Juneau can maintain its economic benefits from tourism while effectively alleviating environmental pressure and improving resident satisfaction, achieving sustainable tourism development.

Sincerely,
Research Team

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